A Descriptive Bootstrap Trend for Accuracy Deterioration Across Labeled Batches

Jabulani Monchwe
Applied Business Analytics and Decision Making
North-West University
IDEA-Research Club Chairperson
Johannesburg, South Africa
monchwej.25@gmail.com

Martin Chanza
Applied Business Analytics and Decision Making
North-West University
Mafikeng, South Africa
martin.chanza@nwu.ac.za

Olorato Motitswane
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
oloratomotitswane@gmail.com

Mfundo Monchwe
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
diditmfundo@gmail.com

***Abstract*—Once outcomes arrive after deployment, performance can be summarized over dated batches. This paper documents ReliableML's deterioration analysis: accuracy is calculated for each labeled batch, regressed on time, and the slope is bootstrapped by resampling rows within each batch. A deterioration flag is raised only when the percentile interval for the accuracy slope lies entirely below zero. The procedure is deliberately described as a descriptive trend analysis. It does not adjust for case mix, serial dependence, changes in labeling, or varying batch precision and cannot attribute a decline to a particular cause.**

***Index Terms*—model monitoring, performance deterioration, accuracy trend, bootstrap, concept drift**

I. Introduction

Input drift may precede, accompany, or occur without a change in predictive performance. Detecting an accuracy decline requires time-indexed true outcomes, which are often delayed in production. Concept-drift literature emphasizes that changes in populations and target relationships vary across settings and require context-specific treatment [1]. ReliableML's method summarizes a simple temporal association [3]; it is not a drift-adaptation algorithm or a causal diagnosis.

II. Estimand and model

Let batch $j = 1, \ldots, J$ have timestamp t_j, true labels y_{ij}, and predictions $\hat{y}_{ij}$. Its empirical accuracy is

$$a_j = \frac{1}{n_j} \sum_{i=1}^{n_j} \mathbb{I}(y_{ij} = \hat{y}_{ij}).$$

After ordering batches by timestamp, ordinary least squares fits $a_j = \beta_0 + \beta_1 t_j + \varepsilon_j$. The slope estimate is

$$\hat{\beta}_1 = \frac{\sum_j (t_j - \bar{t})(a_j - \bar{a})}{\sum_j (t_j - \bar{t})^2}.$$

The estimate is measured as accuracy change per unit of the timestamp representation. The report also provides the regression p-value and requires at least three distinct dates. Each batch contributes one equally weighted observation regardless of sample size; the point estimate is therefore not pooled per-row accuracy and does not weight by precision.

III. Within-batch bootstrap

For each bootstrap replicate b, the algorithm independently samples n_j row indices with replacement within every batch j, recalculates $a_j^{(b)}$, and refits the regression to obtain $\hat{\beta}_1^{(b)}$, following the general bootstrap principle [2]. The reported percentile interval is

$$\left[Q_{\alpha/2}\{\hat{\beta}_1^{(b)}\}, Q_{1-\alpha/2}\{\hat{\beta}_1^{(b)}\} \right],$$

where $1-\alpha$ is the configured confidence level. A deterioration flag is set only when the upper endpoint is below zero. The bootstrap reflects within-batch row resampling variation under an independent-row assumption; it does not resample time points or estimate uncertainty from the choice of time window.

IV. Data requirements and operational protocol

Every record must contain a consistent numeric or parseable date/time timestamp, aligned true and predicted labels, and a distinct date. Three dates are the minimum accepted by the implementation, but more periods are usually needed for a

credible trend analysis. Freeze prediction and label definitions, document outcome delay and missingness, and retain batch sizes. Analyze performance separately by important groups and inspect calibration and class-specific errors in addition to accuracy.

V. Assumptions and limitations

The linear trend is an approximation that can obscure nonlinear changes, abrupt breakpoints, seasonality, autocorrelation, or recovery. A declining slope may reflect changing case mix, population composition, policy, data collection, or labeling rather than model deterioration. Unequal batch sizes receive equal weight in the fitted line, while a small batch's accuracy is noisier than a large batch's. Within-batch row bootstrap does not handle repeated users, facilities, or time dependence. The ordinary regression p-value inherits its own assumptions and is not a causal test. A negative slope interval is evidence of a downward pattern under this design, not a trigger that automatically justifies retraining.

VI. Conclusion

ReliableML provides a transparent first-order summary for dated labeled batches: estimated accuracy slope, row-bootstrap interval, regression p-value, and a conservative interval-based flag. Sound operational conclusions require richer temporal designs, adequate batch sizes, stable labels, and adjustment or stratification for relevant population changes.

References

[1] J. Gama, I. Zliobaite, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, 2014.

[2] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[3] J. Monchwe, O. Motitswane, and M. Monchwe, *ReliableML*, software, version 0.1.1, 2026. https://github.com/Jabualani/ML-Reliabilty.